DIGITAL BE PROJECT LOG BOOK

PDF Pipeline Audit

Defects, severity, and remediation plan for the final logbook PDF.

7	12	5
RELEASE BLOCKERS	HIGH SEVERITY	MEDIUM SEVERITY

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27 October 2026 Read-only audit | no application files changed Commit 6c4ca62

1. Executive summary

The PDF pipeline has the right architecture: one set of sheet components, a server-side gate, and Puppeteer rendering the same `/print/[groupId]` route used on screen. The *rendered document*, however, is not release-ready. Pagination is undefined, long content is clipped without warning, signature names are not stored, and final PDFs are delivered through a public URL.

> **Verdict:** do not ship the official PDF until P0-1 through P0-7 are fixed. The three structural defects (page breaks, flow-page continuation, fixed-height clipping) share one root cause and should be fixed together.

7	12	5	24
P0 - BLOCKERS	P1 - HIGH	P2 - MEDIUM	TOTAL FINDINGS

Top five defects

1. **Page breaks are defeated.** Every printed sheet is wrapped in its own `.paper-desk`, so `.paper-page:last-child` matches every sheet and disables the forced break.
2. **Long sections are silently clipped.** Sheets are fixed at one A4 height with `overflow: hidden`; long reviews, competitions, papers, and notes lose content.
3. **Activity pages print a false lock notice.** The PDF passes `editable=false`, which triggers the "charts unlock once the HOD approves" message on every activity page.
4. **A session failure produces a PDF of the login screen.** The Puppeteer navigation result is never checked for status or final URL.
5. **Final PDFs are public.** Files under `/media/pdf/...` are served without authentication; names are predictable from the group number and a millisecond timestamp.

What the PDF looks like today

PAGINATION
A short RTM and the Cost Estimation sheet can begin on the same physical page, with the second college header halfway down.

CONTENT
A 1,000-character review remark pushes the reviewer signature block past the sheet; it is clipped, not continued.

NUMBERING
The document has no global page numbers; members and schedule each restart at "Page 1 of 2".

SECURITY
Anyone with the generated URL can download the full logbook without signing in.

2. Release blockers (P0)

ID	Severity	Finding, evidence, and fix
P0-1	BLOCKER	**Forced page breaks are disabled** Every sheet component returns its own `.paper-desk` wrapper in print mode. Each `.paper-page` is therefore the last child of its wrapper, so `.paper-page:last-child { break-after: auto }` matches every sheet. Sections are laid out as one continuous strip and split at arbitrary page boundaries. **Evidence:** `app/print/[groupId]/page.tsx:149`; `components/activity/activity-chart-sheet.tsx:274`; `app/globals.css:311-321`. **Fix:** remove the per-sheet wrapper, keep one print root, and apply page breaks to the flat page list.

ID	Severity	Finding, evidence, and fix
P0-2	BLOCKER	**Flow sheets merge and lose the header on continuation** RTM, Cost, Notes, and sponsored meetings use `flow` (height auto). Only the first page receives the college header; a long sheet continues onto an unheaded page, and a short one lets the next section start below it on the same page. **Evidence:** `components/paper/paper-page.tsx:49`; `app/globals.css:116-120`. **Fix:** paginate flow content into explicit `PaperPage` chunks with a repeated header and page number.
P0-3	BLOCKER	**Long sections are silently clipped** Fixed-height sheets have `overflow: hidden`. A 17-question review with long remarks, a long competition or paper list, or a long rules acknowledgement can exceed A4 and lose official content. The gate can still declare the group complete. **Evidence:** `app/globals.css:97-110`; `components/reviews/review-sheet.tsx:238-299`. **Fix:** chunk rows and free text into continuation sheets; assert the final row and signature in extracted PDF text.
P0-4	BLOCKER	**Every activity page prints a false lock notice** The print route passes `editable={false}`. The notice condition is `(mode === "student" || mode === "guide") && !editable`, so it is always true in the PDF. Each activity page tells the reader the charts are locked. **Evidence:** `components/activity/activity-chart-sheet.tsx:249-252`; `app/print/[groupId]/page.tsx:196-209`. **Fix:** gate the notice on `!print` or on the group status.
P0-5	BLOCKER	**The PDF can be the login or error screen** `page.goto()` is not checked for status, final URL, or a readiness marker. `requireUser()` redirects to `/login`; `canViewGroup` redirects to the role home. Both return a 200 HTML page, which Puppeteer prints as the logbook. **Evidence:** `lib/services/pdf.ts:91-101`; `app/print/[groupId]/page.tsx:64-76`. **Fix:** assert `response.ok()`, the final pathname, and a `data-pdf-ready` marker.
P0-6	BLOCKER	**Final PDFs are publicly downloadable** Files are saved under `/media/pdf/...`. The media route has no authentication and is outside the middleware matcher. Filenames combine the group number with `Date.now()`, so the URL is guessable. The document contains photos, phone numbers, marks, and signatures. **Evidence:** `lib/services/pdf.ts:116-121`; `app/media/[...path]/route.ts:24-39`; `middleware.ts:28`. **Fix:** require authentication or a signed token, use random filenames, revoke on permission change, and send `Cache-Control: private, no-store`.
P0-7	BLOCKER	**A4 geometry mismatch can create sliver pages** The sheet is `794 x 1123px`; true A4 at 96dpi is `793.70 x 1122.52px`. With `@page { margin: 0 }` and `preferCSSPageSize`, the 0.3/0.5px overflow can produce blank or sliver pages. **Evidence:** `app/globals.css:80-83`; `components/paper/paper-page.tsx:24-27`; `lib/services/pdf.ts:97-101`. **Fix:** use exact `210mm x 297mm` in print media and add `break-inside: avoid`.

3. High-severity defects (P1)

ID	Severity	Finding, evidence, and fix
P1-8	HIGH	**RTM/Cost line count is wrong** `lines` is derived from `text.split("\n").length`. A 2,000-character paragraph without newlines gets five ruled lines; the read-only overlay overflows over the "Attached file" section. **Evidence:** `components/records/content-sheet.tsx:94`; `components/paper/field.tsx:57-63`. **Fix:** compute lines from characters per line plus newlines, or paginate the block; add `overflow-wrap: anywhere`.

ID	Severity	Finding, evidence, and fix
P1-9	HIGH	**Long fields overlap or clip** Read-only multiline fields are absolute overlays with no clipping. Review remarks allow 1,000 characters in a single-line cell; a tall row pushes the sheet past A4. Notes and Remarks/Suggestions have the same problem. **Evidence:** `components/paper/field.tsx:51-90`; `lib/validation/review.ts:42`. **Fix:** give text blocks deterministic capacity and spill into continuation sheets.
P1-10	HIGH	**No document-wide page numbers** Only members and schedule pass a page number. The PDF has two separate "Page 1 of 2" series and no numbering on the remaining pages. **Evidence:** `components/student/members-sheet.tsx:103`; `components/schedule/schedule-sheet.tsx:76`. **Fix:** build a page plan in the print route and pass numbering to every page, including continuations.
P1-11	HIGH	**Signature identities are not snapshotted** The schema stores IDs and timestamps, not display names. Names are resolved live from `User`, or reconstructed from `AuditLog` for topic scores, marks submission, and sponsored student signs. Renames rewrite history; deletions render a dash; review fallback can attribute an old signature to the current guide. Sponsored meetings store no signer IDs at all. **Evidence:** `prisma/schema.prisma:225-249, 318-388, 462-480`; `lib/services/topic.ts:90-131`; `lib/services/evaluation.ts:140-181`. **Fix:** store signer id, display name, and timestamp on the row at sign time.
P1-12	HIGH	**Signature times use the server time zone** `toLocaleString("en-GB")` has no `timeZone`. Render runs UTC, so signatures show UTC; an Indian browser shows IST, causing hydration differences. **Evidence:** `lib/format.ts:3-14`. **Fix:** use `COLLEGE_TIME_ZONE ?? "Asia/Kolkata"` in both formatters.
P1-13	HIGH	**Concurrent generation duplicates and races** Two clicks both pass the gate and render. Each writes a file, creates a `PdfExport` row and audit entry, and races on `group.status`. The GET route has side effects, so prefetch or retry can trigger generation. **Evidence:** `lib/services/pdf.ts:108-140`; `app/api/groups/[groupId]/pdf/route.ts:11-24`. **Fix:** per-group lock; reuse the latest export or return 409; make generation POST.
P1-14	HIGH	**Orphan file if the DB write fails** `writeFile` happens before the transaction. A DB failure leaves a file with no `PdfExport` row and no audit trail. **Evidence:** `lib/services/pdf.ts:120-138`. **Fix:** write to a temp file, commit the transaction, then rename; delete on failure.
P1-15	HIGH	**PDFs are stored only on local disk** Even with `UPLOAD_PROVIDER=supabase`, PDFs go to `uploadRoot()`. Render's filesystem is ephemeral; a redeploy or another instance makes the saved URL 404. **Evidence:** `lib/services/pdf.ts:116-121`; `lib/upload.ts`. **Fix:** upload to Supabase Storage when configured.
P1-16	HIGH	**PDF_BASE_URL is ignored** The route uses `NEXTAUTH_URL ?? request.nextUrl.origin`. The fallback comes from the Host header, so a proxy or custom-domain mismatch can make Puppeteer fetch the wrong host and send the session cookie there. **Evidence:** `.env.example:49`; `app/api/groups/[groupId]/pdf/route.ts:18-19`. **Fix:** use `PDF_BASE_URL` first and validate the origin.

ID	Severity	Finding, evidence, and fix
P1-17	HIGH	**PDF generation is slow and can time out** The gate runs the completion engine, then `/print` re-queries everything: about 250 queries, including ~150 sequential upserts that materialize activity and review rows. All inside a 90-second navigation. With `connection_limit=1` this is a real timeout path. **Evidence:** `lib/services/pdf.ts:93-96`; `app/print/[groupId]/page.tsx:78-114`; `lib/services/activity-chart.ts:59-72`. **Fix:** batch materialization; batch print queries; replace `networkidle0` with `load` plus font/image readiness.
P1-18	HIGH	**Every download regenerates; no cache headers** A new PDF is rendered, written, and recorded on every request. The response has no `Cache-Control`. **Evidence:** `lib/services/pdf.ts:113-121`; `app/api/groups/[groupId]/pdf/route.ts:25-30`. **Fix:** store a content version/hash and reuse the latest matching export; send `Cache-Control: private, no-store`.
P1-19	HIGH	**Gate and render are not atomic** The gate checks completion, then the print page re-reads the database. An unlock, guide reassignment, permission revoke, or edit between the two can produce a PDF containing unapproved data. **Evidence:** `lib/services/pdf.ts:113-114`; `app/print/[groupId]/page.tsx:64-114`. **Fix:** re-check the gate after rendering, or render from a read-only snapshot recorded on `PdfExport`.

4. Medium-severity defects (P2)

ID	Severity	Finding, evidence, and fix
P2-20	MEDIUM	**Workflow text leaks into the official paper** The evaluation sheet prints a `Status` column with raw `APPROVED`. Reviews print "Last unlocked by the HOD: ...". Rejected topics print the reason in red (colour is cover-only). Topic 2 prints "Topic 1 has been approved, so Topic 2 is not required." An empty Notes sheet prints "No notes yet." **Evidence:** `components/evaluation/eval-sheet.tsx:241,270`; `components/reviews/review-sheet.tsx:301-305`; `components/topics/topic-sheet.tsx:173-176`. **Fix:** render workflow state only when `!print`; omit empty optional sheets.
P2-21	MEDIUM	**The gate accepts visibly incomplete sections** Examiner feedback counts as complete with answers only - no internal/external names, remarks, or signature. One competition row counts without a certificate; one paper row counts without an upload. Sponsor feedback is complete without `signedAt`. **Evidence:** `lib/completion/sections.ts:175-200, 229-238, 241-274`. **Fix:** require every field the sheet renders, including signatures and uploads.
P2-22	MEDIUM	**Error statuses are wrong** Every `AppError` becomes 403, so a missing group is 403 instead of 404. Render timeouts become a generic 500 after a long wait. **Evidence:** `app/api/groups/[groupId]/pdf/route.ts:31-38`. **Fix:** map error codes explicitly; return 404 for not-found and 502/503 for render failures.
P2-23	MEDIUM	**"Signed by -" on legacy rows** A timestamp without a resolvable name renders an em dash. Combined with the live-name lookup, this can misattribute or erase a signature in an official document. **Evidence:** `lib/format.ts:28-31`; `components/paper/sign-cell.tsx:32-37`. **Fix:** snapshot names (P1-11) and show a clear unavailable state for legacy rows.

ID	Severity	Finding, evidence, and fix
P2-24	MEDIUM	**No PDF content or visual tests** The existing suite tests only the gate. Nothing checks page count, headers, page numbers, clipping, login-page detection, or signature provenance. **Evidence:** `tests/pdf.test.ts:7-11`. **Fix:** add a Puppeteer fixture render with text extraction and a Playwright screen/print comparison.

5. Remediation plan

The work is separable into four phases. Phase 1 fixes the document structure and should land as one change; splitting it risks a partially paginated PDF.

1 Page structure and paper fidelity

1. Remove per-sheet `paper-desk` print wrappers; keep one print root.
2. Set exact A4 millimetre sizing and flat-list page-break rules; add `break-inside: avoid`.
3. Chunk reviews, evaluations, competitions, papers, rules, and notes into continuation pages; compute RTM/Cost line capacity.
4. Add a document page plan for numbering; remove the local members/schedule numbering.
5. Suppress workflow notices, status columns, unlock reasons, and red rejection text in print.

2 Pipeline reliability and security

1. Validate `page.goto` status, final URL, and a readiness marker.
2. Wire `PDF_BASE_URL`; wait for fonts and images; replace `networkidle0`.
3. Add a generation lock, temp-file/rename, cleanup, and Supabase storage.
4. Protect `/media/pdf` or serve through an authenticated route; random filenames; no-store.
5. Add export reuse and close the gate/render race.

3 Signature provenance (schema approval required)

Add snapshot columns and write them at sign time. No columns are removed; legacy rows fall back to the current relation or audit lookup.

Model	Columns
`ActivityEntry`	`studentConfirmedByName`, `guideVerifiedByName`
`ReviewForm`	`submittedByName`, `approvedByName`
`ReviewScore`	`guideSignedByName`
`TopicProposal`	`scoredById`, `scoredByName`, `scoredAt`
`MarksSheet`	`submittedById`, `submittedAt`, `submittedByName`, `approvedByName`
`SponsoredMeeting`	`studentSignedById`, `studentSignedByName`, `guideSignedById`, `guideSignedByName`, `externalGuideSignedById`, `externalGuideSignedByName`

Why: master rule 9 requires user id, display name, and timestamp. The current code reconstructs topic-score and marks-submission signatures from `AuditLog` and resolves every other name live from `User`.

4 Tests and regression protection

1. Render a complete fixture with Puppeteer; assert page count, header on every page, sequential numbers, and absence of workflow copy.
2. Add a 1,000-character review remark and a long RTM paragraph; assert continuation with header and a present final signature.
3. Rename a signer after signing; delete another; assert the original names survive.
4. Assert unauthenticated `/media/pdf/...` is rejected and a redirect to `/login` fails generation.
5. Extend gate tests for examiner signatures, certificates, and paper uploads.

6. Method, limits, and next step

This is a static code audit of the print route, sheet components, print CSS, PDF service, media route, completion engine, Prisma schema, and tests at commit `6c4ca62` (22 October 2026). No application files were modified. Live rendering was not captured in this environment because the configured database was not reachable; findings are based on the current source.

Release checklist

- [ ] P0-1/P0-2/P0-3: one sheet per page; no clipping; header on every continuation page.
- [ ] P0-4: no workflow notices on any printed sheet.
- [ ] P0-5: generation fails loudly on login/error HTML.
- [ ] P0-6: PDF URL requires authentication and cannot be guessed.
- [ ] P0-7: exact A4; no blank or sliver pages.
- [ ] P1-10: sequential document page numbers.
- [ ] P1-11/P1-12: signature names and times are stable and local.
- [ ] P1-13/P1-14/P1-15: one export per request; no orphans; durable storage.
- [ ] P1-17/P1-18: generation within the timeout; repeat download reuses the export.
- [ ] P2-24: PDF content and visual tests in CI.

Decisions needed before implementation

1. **Signature snapshot schema:** approve the additive columns in section 5, phase 3.
2. **Scope:** fix PDF correctness only, or also fold in the PDF performance work (query batching, export reuse, connection pooling).
3. **Paper policy:** omit a non-required Topic 2 page or print a blank sheet; confirm the two evaluation-committee signatories when the guide both submits and approves.

Recommended next change: implement phase 1 as a single pull request, then add the Puppeteer content test before touching the pipeline. The structural fix changes the page count, so capture any visual baseline after phase 1.

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